

Understanding HTTP Status Codes

Assignment Article

Introduction

HTTP status codes are three-digit numbers sent by a web server to a client, such as a web browser or mobile application, after the client makes a request. They help the client understand what happened to the request.

For example, when a user opens a website, the browser sends a request to the server asking for the webpage. The server processes the request and sends back a response that includes an HTTP status code. The code tells the browser whether the request was successful, whether more action is needed, or whether an error occurred.

HTTP status codes are divided into five main categories:

- 1xx - Informational responses
- 2xx - Successful responses
- 3xx - Redirection responses
- 4xx - Client error responses
- 5xx - Server error responses

Below are two examples from each category, making ten status codes in total.

1xx - Informational Responses

The 1xx category shows that the server has received the request and that processing is continuing.

100 - Continue

What it means: The 100 Continue status code tells the client that the first part of its request has been received and that it can continue sending the rest of the request.

When it is used: It is commonly used when a client wants to send a large amount of information to a server. Before sending the complete data, the client can first check whether the server is willing to accept it.

Real-world example: Imagine a user uploading a very large video file to a website. The browser can first send the request headers to the server. If the server replies with 100 Continue, the browser knows it can proceed with uploading the video.

101- Switching Protocols

What it means: The 101 Switching Protocols status code means that the server has agreed to change from one communication protocol to another.

When it is used: It is often used when a normal HTTP connection needs to be upgraded to another protocol, such as WebSocket.

Real-world example: A live chat application may begin communication using HTTP. The browser may then request a WebSocket connection so that messages can be exchanged instantly. If the server accepts the change, it responds with 101 Switching Protocols.

2xx - Successful Responses

The 2xx category means that the client's request was successfully received, understood, and processed by the server.

200 - OK

What it means: The 200 OK status code means that the request was successful and the server has returned the requested information.

When it is used: It is one of the most common HTTP status codes. It is normally used when a webpage, image, API response, or other resource is successfully retrieved.

Real-world example: When a user visits a news website and the page loads correctly, the server may return 200 OK together with the webpage content.

201 - Created

What it means: The 201 Created status code means that the request was successful and that a new resource has been created on the server.

When it is used: It is commonly used after a POST request that creates new information, such as a new user account, blog post, order, or database record.

Real-world example: If a user creates a new account on an online shopping website, the registration information is sent to the server. After successfully creating the account, the server can respond with 201 Created.

3xx - Redirection Responses

The 3xx category means that the client may need to take another action or go to another location to complete the request.

301 - Moved Permanently

What it means: The 301 Moved Permanently status code means that the requested resource has permanently moved to a new web address.

When it is used: It is commonly used when a website changes its domain name or when the address of a webpage is permanently changed.

Real-world example: Suppose a company changes a webpage from `example.com/old-products` to `example.com/products`. When someone visits the old address, the server can return 301 Moved Permanently and send the browser to the new address. This also tells search engines that the new address should replace the old one.

302 - Found

What it means: The 302 Found status code means that the requested resource is temporarily available at another location.

When it is used: It is used for temporary redirects. Unlike a 301 redirect, the original address is expected to be used again in the future.

Real-world example: An online store may temporarily redirect customers from a normal product page to a special promotional page during a holiday sale. The server can use 302 Found because the redirect is only temporary.

4xx - Client Error Responses

The 4xx category means that there is a problem with the request sent by the client. The problem may be caused by an incorrect web address, missing authentication, invalid information, or lack of permission.

400 - Bad Request

What it means: The 400 Bad Request status code means that the server cannot process the request because something about the request is incorrect or invalid.

When it is used: It can occur when required information is missing, incorrectly formatted, or invalid.

Real-world example: Suppose an application asks a user to submit their age as a number, but the application sends incorrectly formatted information to the server. If the server cannot understand the request, it may respond with 400 Bad Request.

404 - Not Found

What it means: The 404 Not Found status code means that the server was reached successfully, but the requested resource could not be found.

When it is used: It usually appears when a user enters an incorrect URL, follows a broken link, or requests a page that has been deleted.

Real-world example: If a website has a page at `example.com/about` but a user enters `example.com/abot`, the server may respond with 404 Not Found because no page exists at that address. Web developers often create custom 404 pages that help users return to the homepage or search for another page.

5xx - Server Error Responses

The 5xx category means that the server received a valid request but was unable to complete it because something went wrong on the server side.

500 - Internal Server Error

What it means: The 500 Internal Server Error status code means that an unexpected problem occurred on the server and prevented it from completing the request.

When it is used: It may occur because of programming errors, incorrect server configuration, database problems, or other unexpected failures.

Real-world example: A user may click “Place Order” on an online shopping website. If the server application crashes while processing the order, the server may respond with 500 Internal Server Error. The issue normally needs to be investigated by the website developers or server administrators.

503 - Service Unavailable

What it means: The 503 Service Unavailable status code means that the server is temporarily unable to handle the request.

When it is used: It is commonly used when a server is undergoing maintenance, is overloaded with too many requests, or temporarily cannot provide a service.

Real-world example: Imagine that thousands of people attempt to buy concert tickets at exactly the same time. If the website servers become overloaded, some users may receive a 503 Service Unavailable. The website may become available again once the number of requests decreases or the technical problem is fixed.

Summary Table

Category	Status Code	Meaning
1xx	100	Continue
1xx	101	Switching Protocols
2xx	200	OK
2xx	201	Created
3xx	301	Moved Permanently
3xx	302	Found / Temporary Redirect
4xx	400	Bad Request
4xx	404	Not Found
5xx	500	Internal Server Error
5xx	503	Service Unavailable

Conclusion

HTTP status codes are an important part of communication between clients and servers. They allow a server to quickly tell a browser or application what happened after a request was made.

The five categories make the responses easier to understand. 1xx codes provide information, 2xx codes show success, 3xx codes handle redirects, 4xx codes indicate problems with the client's request, and 5xx codes indicate problems on the server.

Understanding these status codes is useful for web developers because they make it easier to build websites, create APIs, identify errors, troubleshoot problems, and provide better experiences for users. In simple terms, HTTP status codes act like short messages between the client and server, helping both sides understand the result of every web request.